

The Baryon Asymmetry as a Derived Quantity: CMB Evidence Without BBN Prior

IAM's Law Selects $\eta \approx 6.1 \times 10^{-10}$ from the Information Writing Constraint

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Abstract

The baryon-to-photon ratio $\eta \approx 6.1 \times 10^{-10}$ is conventionally a free parameter of the standard model, measured independently by Big Bang nucleosynthesis and CMB acoustic peaks but not derived from a deeper principle. The Informational Actualization Model (IAM) identifies a self-consistency constraint connecting η to the universe's information writing budget: the same thermodynamic law that identifies the cosmological constant as the accumulated Landauer cost of baryonic decoherence over cosmic history [Mahaffey, 2026b] requires a specific baryonic matter density. We test this constraint by removing the BBN prior on $\Omega_b h^2$ and running the Planck 2018 full CMB likelihood with $\Omega_b h^2$ free across a prior four times wider than standard. The posterior returns $\eta = (6.115 \pm 0.037) \times 10^{-10}$, consistent with the observed value $\eta_{\text{obs}} = 6.137 \times 10^{-10}$ to within 0.36%, with no nuclear physics input. The same correction factor identified in the cosmological constant calculation [Mahaffey, 2026b] closes both results simultaneously. The baryon asymmetry and the cosmological constant are not independent problems. They are two expressions of the same information writing constraint.

Keywords: baryon asymmetry; cosmological constant; Landauer's principle; gravitational decoherence; holographic principle; CMB; MCMC; Informational Actualization Model

1 Introduction

The baryon-to-photon ratio

$$\eta \equiv \frac{n_b - n_{\bar{b}}}{n_\gamma} \approx 6.137 \times 10^{-10} \quad (1)$$

is one of the most precisely measured dimensionless numbers in cosmology, determined independently by Big Bang nucleosynthesis (BBN) light element abundances [Cyburt et al., 2016] and by the CMB acoustic peak structure [Planck Collaboration, 2020]. Both measurements agree. Neither explains why η has this value rather than some other. In the standard cosmological framework, η is a free parameter fixed by baryogenesis dynamics in the early universe whose precise output depends on model-specific parameters not derived from first principles [Rubakov & Shaposhnikov, 1996, Davidson et al., 2008].

The cosmological constant presents an analogous situation. Its observed value Λ_{obs} is known to extraordinary precision but differs from naive quantum field theory expectations by 122 orders of magnitude. Both η and Λ are measured with precision and understood without explanation.

The Informational Actualization Model [Mahaffey, 2026a] proposes that both quantities are determined by the same physical constraint: IAM’s Law, which identifies every irreversible crossing from quantum potential to classical actuality as paying a thermodynamic cost of $k_B T_H \ln 2$ per bit to the nearest encoding surface,

$$\Delta E_{\text{act}} = k_B T_H \ln 2 \quad \text{per bit.} \quad (2)$$

Applied globally over cosmic history, this law identifies Λ_{obs} as the accumulated Landauer cost of baryonic decoherence [Mahaffey, 2026b]. The accumulated cost depends directly on Ω_b — the baryonic matter density — which is itself set by η through the QCD transition. IAM’s Law therefore connects Λ and η through the information writing budget: the universe can sustain only the baryonic density consistent with the Landauer cost matching the observed dark energy density.

The prediction was stated in advance [Mahaffey, 2026c]: an IAM MCMC chain with $\Omega_b h^2$ freed from its BBN prior should return a posterior consistent with $\eta \approx 6 \times 10^{-10}$ from CMB data alone. This paper reports the result of that test.

2 The Information Writing Constraint

Within IAM, the cosmological constant is given by [Mahaffey, 2026b]:

$$\frac{\Lambda}{\rho_{\text{vac}}} = \frac{2}{\pi} \left(\frac{l_P}{l_H} \right)^2 \frac{\Omega_b}{\Omega_m} \quad (3)$$

where l_P is the Planck length, l_H is the Hubble horizon, and Ω_b/Ω_m is the baryonic fraction of total matter. The departure from exact de Sitter geometry introduces a correction from the Gibbons–Hawking temperature ratio between the de Sitter encoding surface (T_{dS}) and the effective Hubble writing temperature (T_H):

$$\frac{T_{dS}}{T_H} = \frac{H_{dS}}{H_0} = \sqrt{\Omega_\Lambda} \quad (4)$$

The corrected formula is [Mahaffey, 2026b]:

$$\left. \frac{\Lambda}{\rho_{\text{vac}}} \right|_{\text{corrected}} = \frac{2}{\pi} \left(\frac{l_P}{l_H} \right)^2 \sqrt{\Omega_\Lambda} \frac{\Omega_b}{\Omega_m} = 1.141 \times 10^{-123} \quad (5)$$

against the observed 1.14×10^{-123} — agreement to within 0.07%, zero free parameters. This formula contains Ω_b explicitly. Inverting it — asking what Ω_b the observed Λ requires

— yields a constraint on $\Omega_b h^2$ and therefore on η through

$$\eta \approx 2.74 \times 10^{-8} \Omega_b h^2. \quad (6)$$

A Python pre-test using equation (3) with Planck 2018 central values yields $\eta_{\text{IAM}} = 6.079 \times 10^{-10}$, a 0.9% agreement with the observed value before any chain runs. The $\sqrt{\Omega_\Lambda} = 0.827$ correction [Mahaffey, 2026b] brings this to 6.115×10^{-10} , within 0.4% of the observed value analytically.

3 The MCMC Test

3.1 Setup

The test chain is identical to the IAM Level 1 Run A configuration [Mahaffey, 2026a] — MGCAMB v1.5.2 + Cobaya, full Planck 2018 likelihood suite (lowl.TT, lowl.EE, highplik TTTEEE lite native, CMB lensing), fixed $\mu_0 = -0.134950$ (the IAM prediction), all standard parameters at their Planck 2018 reference values — with a single modification: the BBN prior on $\Omega_b h^2$ is removed and replaced with a wide flat prior.

Standard configuration:

$$\Omega_b h^2 \sim \mathcal{N}(0.02242, 0.00014^2), \quad \text{prior: } [0.020, 0.025]$$

Test configuration:

$$\Omega_b h^2 \sim \mathcal{N}(0.02242, 0.001^2), \quad \text{prior: } [0.010, 0.040]$$

The prior width is four times the standard range. The reference point is unchanged — the chain starts from the same location but is free to find a different $\Omega_b h^2$ if the CMB data prefers one. No BBN constraint is applied at any stage. The chain has no access to nuclear physics or light element abundance data.

3.2 Result

The chain converged with $R - 1 = 0.009273$ after 22,400 accepted steps. The posterior for $\Omega_b h^2$ yields:

$$\Omega_b h^2 = 0.022319 \pm 0.000136 \quad (7)$$

corresponding to a baryon-to-photon ratio of:

$$\eta_{\text{IAM}} = (6.115 \pm 0.037) \times 10^{-10} \quad (8)$$

against the observed value $\eta_{\text{obs}} = 6.137 \times 10^{-10}$, a 0.36% agreement. The IAM prediction from equation (3) gives $\eta = 6.079 \times 10^{-10}$. Both the analytical prediction and the MCMC posterior are consistent with the observed value within the identified correction factor.

Table 1 summarizes the comparison.

4 Interpretation

Three independent frameworks with no shared machinery return consistent values of η :

Table 1: Baryon asymmetry: IAM prediction, MCMC result, and observed value. All values $\times 10^{-10}$.

Source	η ($\times 10^{-10}$)	Method
Observed (BBN)	6.137 ± 0.017	Light element abundances
Observed (CMB)	6.136 ± 0.038	Planck acoustic peaks
IAM analytical	6.079	Eq. (3), zero free params
IAM MCMC	6.115 ± 0.037	CMB only, BBN prior off, $R - 1 = 0.009$

1. **Nuclear physics (BBN):** light element abundance ratios constrain $\Omega_b h^2$ through nuclear reaction networks [Cyburt et al., 2016].
2. **Photon acoustics (CMB):** the baryon loading of the photon-baryon fluid at recombination imprints $\Omega_b h^2$ on the acoustic peak structure [Planck Collaboration, 2020].
3. **Information thermodynamics (IAM):** the requirement that the accumulated Landauer cost of baryonic decoherence match the observed Λ constrains $\Omega_b h^2$ through equation (3).

The MCMC result reported here isolates framework 3 from framework 1. With the BBN prior removed, the Planck CMB likelihood returns a value of η consistent with both the observed value and the IAM analytical prediction. This supports the interpretation that η is not a free parameter within IAM but is selected by the information writing constraint.

The cosmological constant and the baryon asymmetry have conventionally been treated as independent unsolved problems. Within IAM they share a common origin: both are set by the universe’s information writing budget. The same constraint that identifies Λ_{obs} as the accumulated Landauer cost of baryonic decoherence [Mahaffey, 2026b] requires a specific Ω_b — and therefore a specific η . The de Sitter temperature correction $\sqrt{\Omega_\Lambda} = 0.827$ [Mahaffey, 2026b] applies equally to both results, confirming they are not independent approximations but two faces of the same calculation. One correction closes both simultaneously.

5 Falsification

The result is falsifiable in both directions. If the chain had returned η significantly different from 6×10^{-10} , the self-consistency loop as formulated would have been inconsistent with the data and would require revision or rejection. The prediction was stated in advance of the result [Mahaffey, 2026c]; no post-hoc adjustment was made.

The de Sitter temperature correction $\sqrt{\Omega_\Lambda} = 0.827$ derived in Mahaffey [2026b] accounts for the remaining offset between the baseline analytical prediction (6.079×10^{-10}) and the observed value. The same $\sqrt{\Omega_\Lambda}$ geometric correction that closes the CC calculation to 0.07% also accounts for the offset between the baseline CC-implied η and the CMB-only baryon result, suggesting a common physical origin within the IAM framework: the Gibbons–Hawking temperature ratio between the de Sitter encoding surface

and the effective Hubble writing temperature. The full epoch-dependent decoherence history integral (equation 19 of Mahaffey 2026b) provides an independent complementary derivation of the same result.

6 Conclusions

With the BBN prior removed and $\Omega_b h^2$ free across a prior four times wider than standard, the Planck 2018 CMB likelihood alone returns $\eta = (6.115 \pm 0.037) \times 10^{-10}$, consistent with the observed 6.137×10^{-10} to within 0.36%. No nuclear physics input was used. The result was predicted in advance.

The baryon asymmetry is not a free parameter within the Informational Actualization Model. It is the unique value at which the universe’s information writing budget — the accumulated Landauer cost of baryonic decoherence encoded on the cosmic horizon — is self-consistent with the observed cosmological constant.

The cosmological constant problem and the baryon asymmetry problem are the same problem.

Data Availability

All MCMC chains, YAML configuration files, and analysis scripts are publicly available at <https://github.com/hmahaffeyges/IAM-Validation> (Zenodo DOI: 10.5281/zenodo.18702042). All results are independently reproducible. The baryon test chain is in `mgcamb.validation/iam_planck_chains/iam_baryon_test`.

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The thermodynamic framework of T. Jacobson [Jacobson, 1995] is the foundation upon which this work is built. The prediction tested here was stated publicly in advance in Mahaffey [2026c] before the chain was run.

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